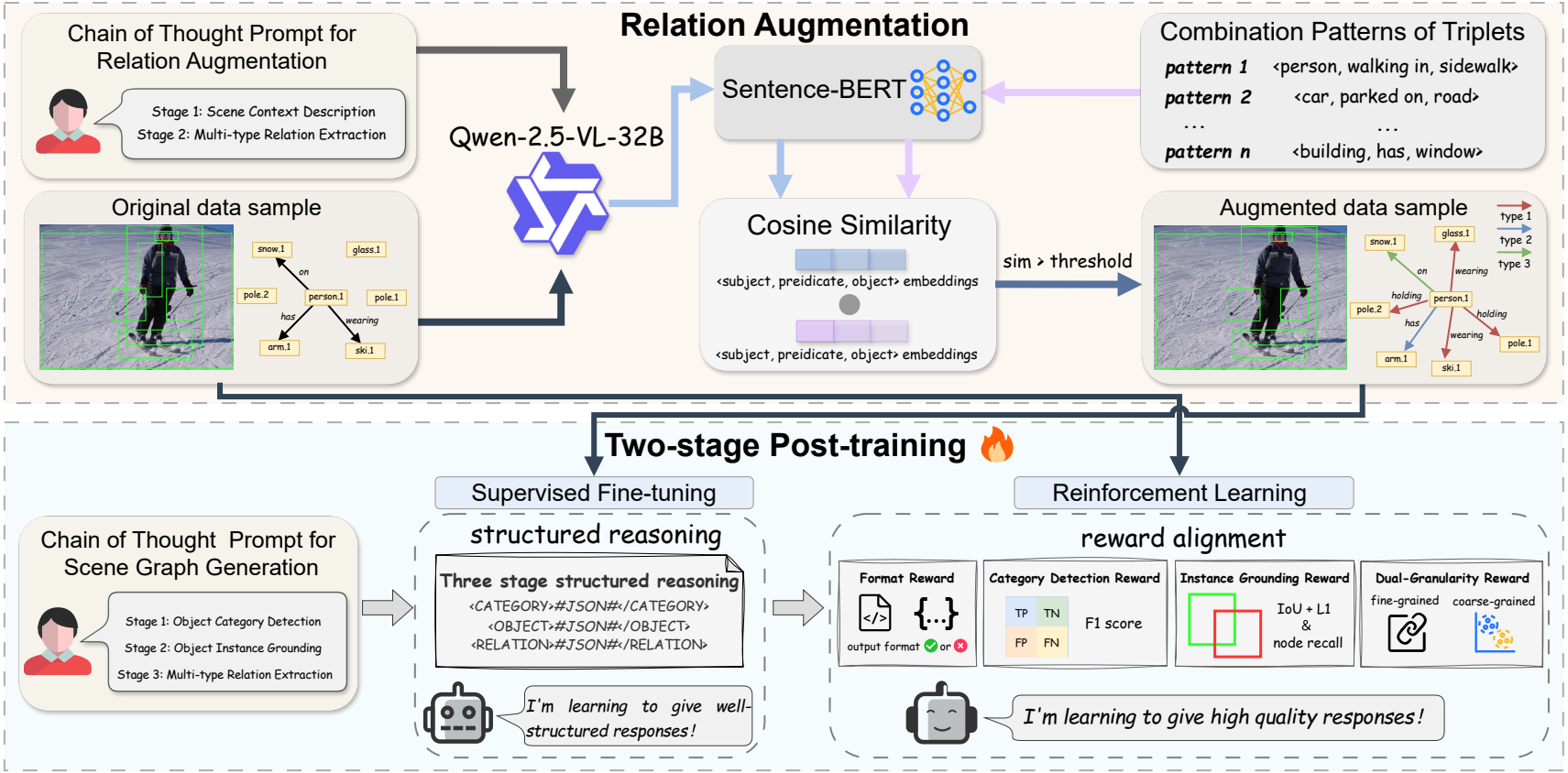
Relation Augmentation
Chain of Thought Prompt for Relation Augmentation
Stage 1: Scene Context Description
Stage 2: Multi-type Relation Extraction
Qwen-2.5-VL-32B
Sentence-BERT
Combination Patterns of Triplets
pattern 1 <person, walking in, sidewalk>
pattern 2 <car, parked on, road>
...
pattern n <building, has, window>
Original data sample
snow.1
glass.1
on
pole.2
person.1
pole.1
has
wearing
arm.1
ski.1
Cosine Similarity
<subject, preidicate, object> embeddings
<subject, preidicate, object> embeddings
sim > threshold
Augmented data sample
type 1
type 2
type 3
holding
Two-stage Post-training
Supervised Fine-tuning
Reinforcement Learning
Chain of Thought Prompt for Scene Graph Generation
Stage 1: Object Category Detection
Stage 2: Object Instance Grounding
Stage 3: Multi-type Relation Extraction
structured reasoning
Three stage structured reasoning
<CATEGORY>#JSON#</CATEGORY>
<OBJECT>#JSON#</OBJECT>
<RELATION>#JSON#</RELATION>
I'm learning to give well-structured responses!
reward alignment
Format Reward
output format or
Category Detection Reward
TP
TN
FP
FN
F1 score
Instance Grounding Reward
IoU + L1
&
node recall
Dual-Granularity Reward
fine-grained
coarse-grained
I'm learning to give high quality responses!